

Relational Model in DBMS

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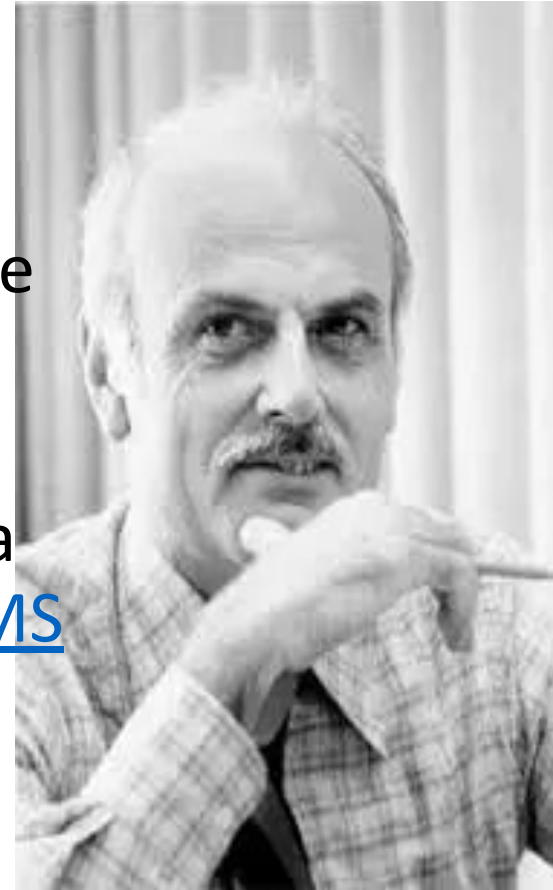


What is the Relational Model?

- The Relational model represents how data is stored in Relational Databases. A relational database consists of a collection of tables, each of which is assigned a unique name.
- Consider a relation STUDENT with attributes ROLL_NO, NAME, ADDRESS, PHONE, and AGE shown in the table.

RDBMS

- E.F. Codd proposed the relational Model to model data in the form of relations or tables.
- After designing the conceptual model of the Database using [ER diagram](#), we need to convert the conceptual model into a relational model which can be implemented using any [RDBMS](#) language like Oracle SQL, MySQL, etc.
- The relational model uses a collection of tables to represent both data and the relationships among those data.
- Each table has multiple columns, and each column has a unique name. Tables are also known as relations.



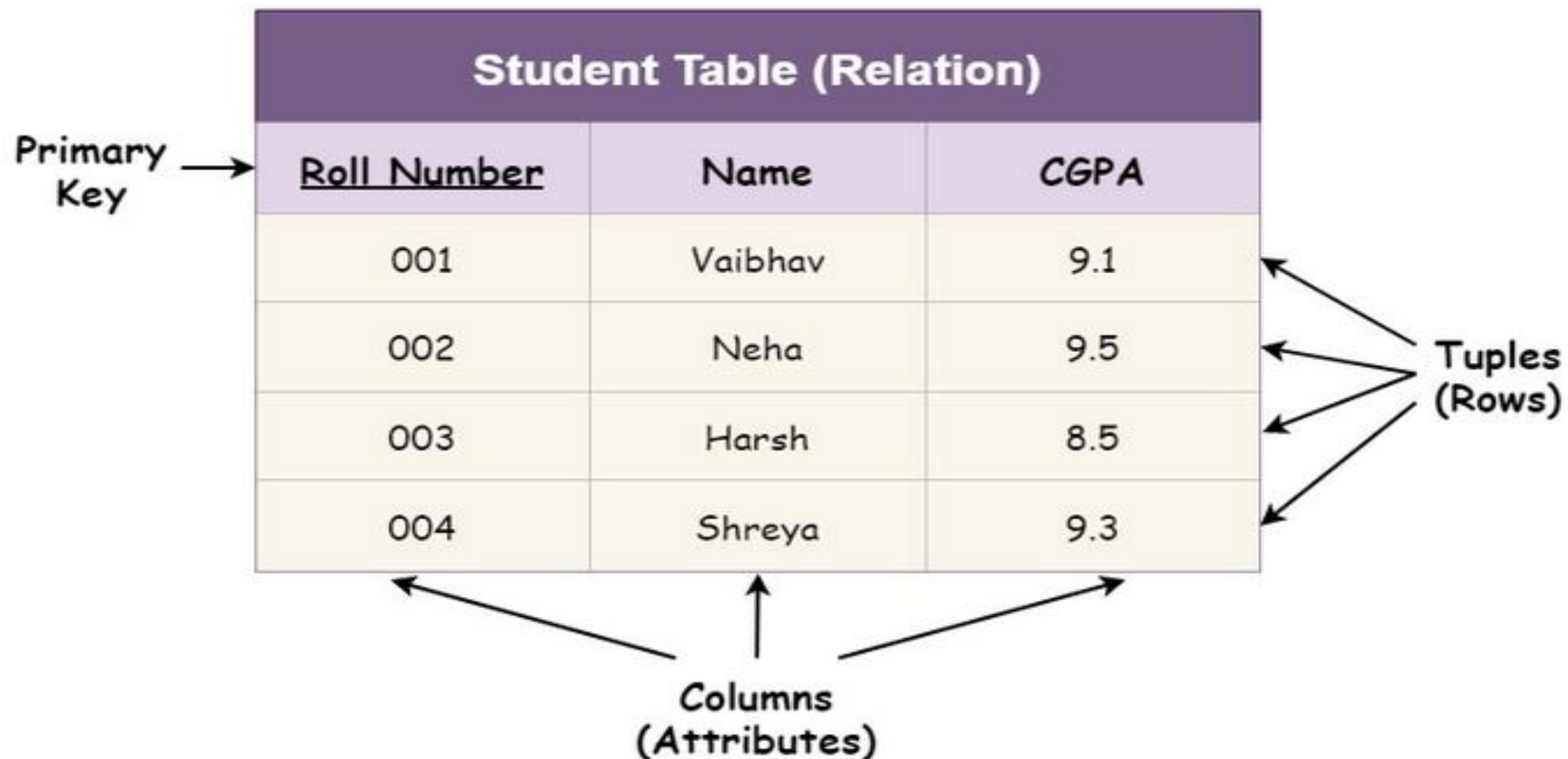
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- The relational data model is the most widely used data model, and a vast majority of current database systems are based on the relational model.
- The relational model is an example of a record-based model.
- Record-based models are so named because the database is structured in fixed-format records of several types.
- Each table contains records of a particular type.
- Each record type defines a fixed number of fields, or attributes.

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- Relational model can represent as a table with columns and rows. Each row is known as a tuple. Each table of the column has a name or attribute.
- **Domain:** It contains a set of atomic values that an attribute can take.
- **Attribute:** It contains the name of a column in a particular table. Each attribute A_i must have a domain, $\text{dom}(A_i)$
- **Relational instance:** In the relational database system, the relational instance is represented by a finite set of tuples. Relation instances do not have duplicate tuples.
- **Relational schema:** A relational schema contains the name of the relation and name of all columns or attributes.
- **Relational key:** In the relational key, each row has one or more attributes. It can identify the row in the relation uniquely.
- **Relational Instance:** It is the collection of records present in the relation at a given time.

Relational Model in DBMS



Example: STUDENT Relation

NAME	ROLL_NO	PHONE_NO	ADDRESS	AGE
Ram	14795	7305758992	Noida	24
Shyam	12839	9026288936	Delhi	35
Laxman	33289	8583287182	Gurugram	20
Mahesh	27857	7086819134	Ghaziabad	27
Ganesh	17282	9028 9i3988	Delhi	40

- In the given table, NAME, ROLL_NO, PHONE_NO, ADDRESS, and AGE are the attributes.
- The instance of schema STUDENT has 5 tuples.
- Degree: It refers to the total number of attributes that are there in the relation. The STUDENT relation defined here has degree 5.

Relational Model Concepts

- A Relational database is based on the relational model. This database consists of various components based on the relational model. These include:
- **Relation:** Two-dimensional table used to store a collection of data elements.
- **Tuple:** Row of the relation, depicting a real-world entity.
- **Attribute/Field:** Column of the relation, depicting properties that define the relation.
- **Attribute Domain:** Set of pre-defined atomic values that an attribute can take i.e., it describes the legal values that an attribute can take.
- **Degree:** It is the total number of attributes present in the relation.
- **Cardinality:** It specifies the number of entities involved in the relation i.e., it is the total number of rows present in the relation.
- **Relational Schema:** It is the logical blueprint of the relation i.e., it describes the design and the structure of the relation. It contains the table name, its attributes, and their types:
- TABLE_NAME(ATTRIBUTE_1 TYPE_1, ATTRIBUTE_2 TYPE_2, ...)
- For our Student relation example, the relational schema will be:
- STUDENT(ROLL_NUMBER INTEGER, NAME VARCHAR(20), CGPA FLOAT)

Properties of Relations

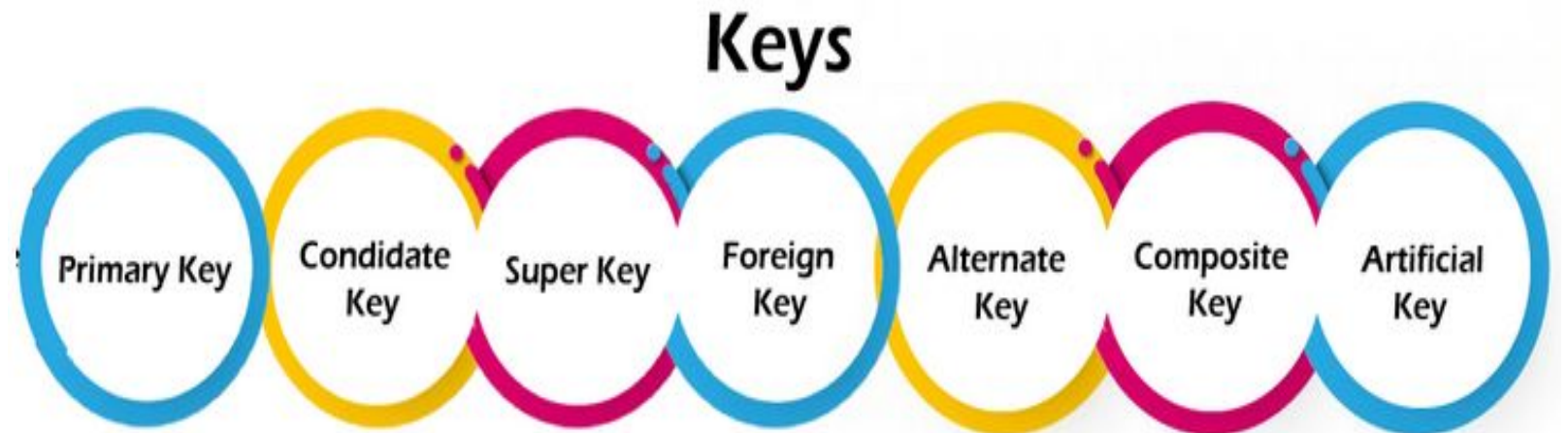
- Name of the relation is distinct from all other relations.
- Each relation cell contains exactly one atomic (single) value
- Each attribute contains a distinct name
- tuple has no duplicate value
- Order of tuple can have a different sequence
- All of the values present in a column hold the same data type
- The columns sequence is not significant

Relation Key

- Relation Key: It is an attribute or a group of attributes that can be used to uniquely identify an entity in a table or to determine the relationship between two tables. Relation keys can be of 6 different types:

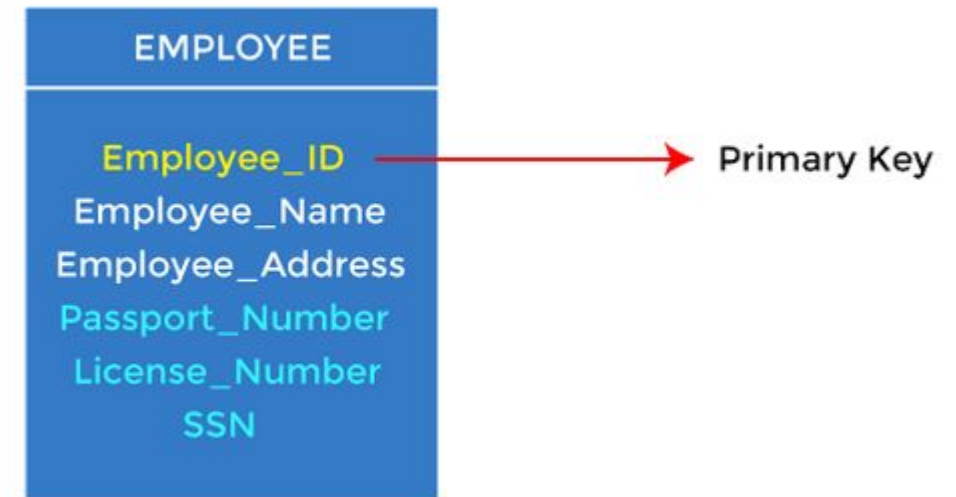
- Primary Key
- Candidate Key
- Super Key
- Composite Key
- Alternate Key
- Foreign Key

Types of keys:



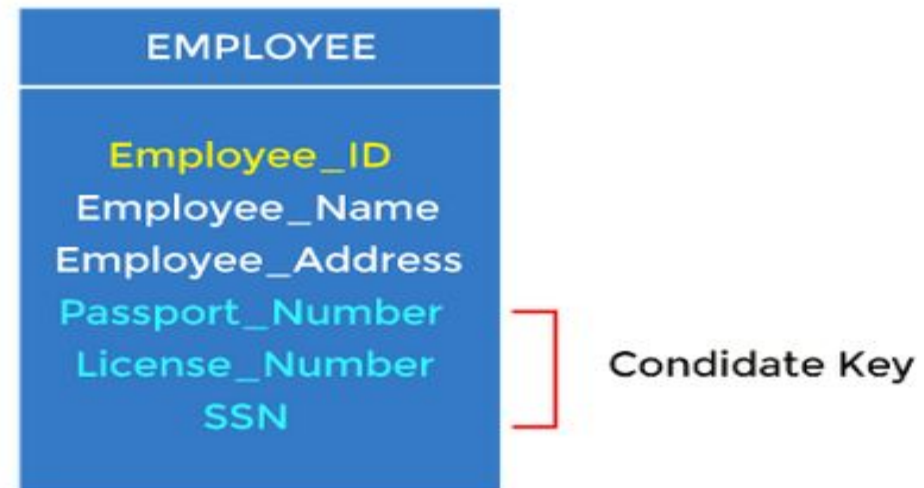
Primary key

- It is the first key used to identify one and only one instance of an entity uniquely. An entity can contain multiple keys. The key which is most suitable from those lists becomes a primary key.
- In the EMPLOYEE table, ID can be the primary key since it is unique for each employee. In the EMPLOYEE table, we can even select License_Number and Passport_Number as primary keys since they are also unique.
- Primary Key is a minimal set of attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.



Candidate key

- A candidate key is an attribute or set of attributes that can uniquely identify a tuple.
- Except for the primary key, the remaining attributes are considered a candidate key.
- The candidate keys are as strong as the primary key.
- **For example:** In the EMPLOYEE table, id is best suited for the primary key. The rest of the attributes, like SSN, Passport_Number, License_Number, etc., are considered a candidate key.

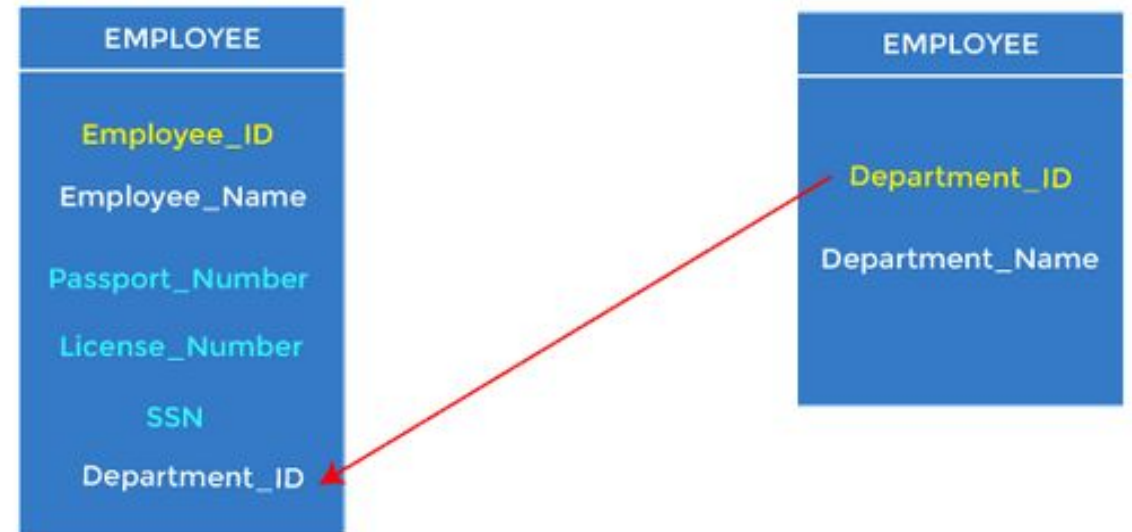


Super Key:

- Super Key is an attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.
- All super keys can't be candidate keys but its reverse is true. In a relation, number of super keys are more than number of candidate keys.

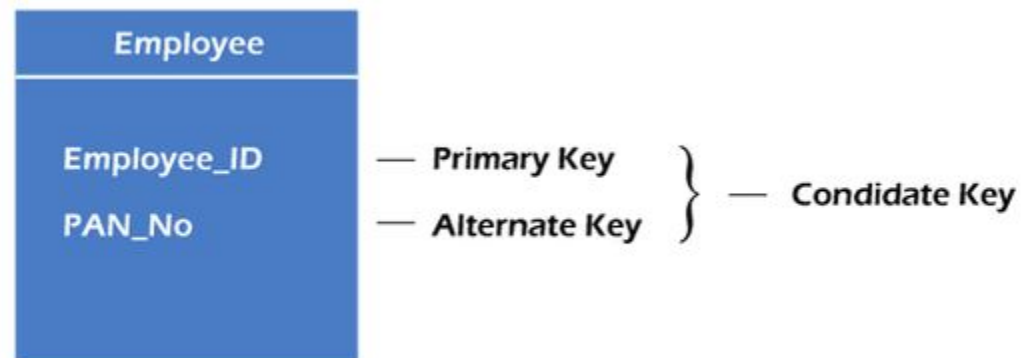
Foreign key

- Foreign keys are the column of the table used to point to the primary key of another table.
- Every employee works in a specific department in a company, and employee and department are two different entities. So we can't store the department's information in the employee table. That's why we link these two tables through the primary key of one table.
- We add the primary key of the DEPARTMENT table, Department_Id, as a new attribute in the EMPLOYEE table.
- In the EMPLOYEE table,
- Department_Id is the foreign key,
- and both the tables are related.



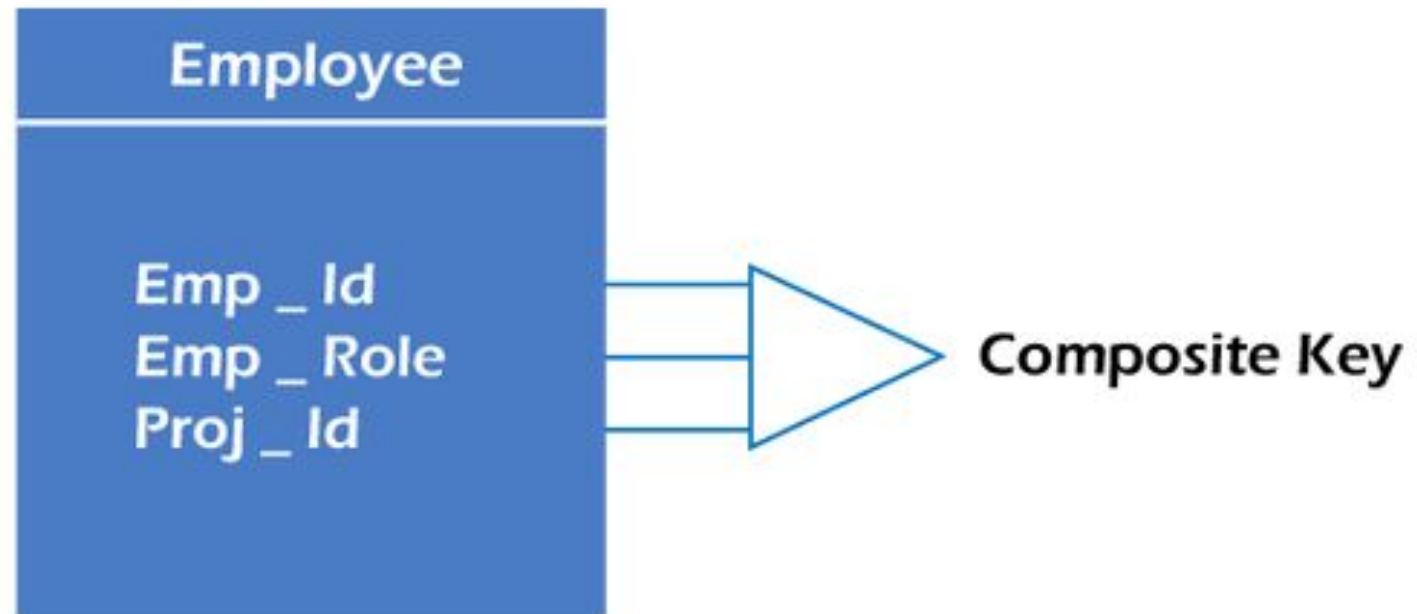
Alternate key

- There may be one or more attributes or a combination of attributes that uniquely identify each tuple in a relation.
- These attributes or combinations of the attributes are called the candidate keys.
- One key is chosen as the primary key from these candidate keys, and the remaining candidate key, if it exists, is termed the alternate key.
- **In other words**, the total number of the alternate keys is the total number of candidate keys minus the primary key. The alternate key may or may not exist. If there is only one candidate key in a relation, it does not have an alternate key.



Composite key

- Whenever a primary key consists of more than one attribute, it is known as a composite key. This key is also known as Concatenated Key.



Artificial key

- The key created using arbitrarily assigned data are known as artificial keys. These keys are created when a primary key is large and complex and has no relationship with many other relations. The data values of the artificial keys are usually numbered in a serial order.



Difference between Super Key and Primary Key:

S.NO	Super Key	Primary Key
1.	Super Key is an attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.	Primary Key is a minimal set of attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.
2.	All super keys can't be primary keys.	Primary key is a minimal super key.
3.	Various super keys together makes the criteria to select the candidate keys.	We can choose any of the minimal candidate key to be a primary key.
4.	In a relation, number of super keys are more than number of primary keys.	While in a relation, number of primary keys are less than number of super keys.
5.	Super key's attributes can contain NULL values.	Primary key's attributes cannot contain NULL values.

Constraints in Relational Model

- Relational models make use of some rules to ensure the accuracy and accessibility of the data.
- These rules or constraints are known as **Relational Integrity Constraints**. These constraints are checked before performing any operation like insertion, deletion, or updation on the data present in a relational database

Different types of Constraints on Relationship model

- **Domain Constraint:** It specifies that every attribute is bound to have a value that lies inside a specific range of values. It is implemented with the help of the Attribute Domain concept.
- **Key Constraint:** It states that every relation must contain an attribute or a set of attributes (Primary Key) that can uniquely identify a [tuple](#) in that relation. This key can never be NULL or contain the same value for two different tuples.
- **Referential Integrity Constraint:** It is defined between two interrelated tables. It states that if a given relation refers to a key attribute of a different or same table, then that key must exist in the given relation.

Anomalies in Relational Model

- When we notice any *unexpected behavior* while working with relational databases, there may be a presence of too much redundancy in the data stored in the database. This can cause [anomalies in the DBMS](#) and it can be of various types such as:
- **Insertion Anomalies:** It is the inability to insert data in the database due to the absence of other data.

For example: Suppose we are dividing the whole class into groups for a project and the *GroupNumber* attribute is defined so that null values are not allowed. If a new student is admitted to the class but not immediately assigned to a group then this student can't be inserted into the database.

Anomalies

- **Deletion Anomalies** - It is the accidental loss of data in the database upon deletion of any other data element.
- **For example:** Suppose, we have an employee relation that contains the details of the employee along with the department they are working in. Now, if a department has one employee working in it and we remove the information of this employee from the table, there will be a loss of data related to the department also. This can lead to *data inconsistency*.

Modification/Update Anomalies

- **Modification/Update Anomalies** - It is the data inconsistency that arises from data redundancy and partial updation of data in the database.

For example: Suppose, while updating the data into the database duplicate entries were entered. Now, if the user does not realize that the data is stored redundantly after updation, there will be data inconsistency in the database.

- All these anomalies can lead to unexpected behavior and inconvenience for the user. These anomalies can be removed with the help of a process known as [normalization](#).

Codd Rules in DBMS

Edgar F. Codd, the creator of the relational model proposed 13 rules known as Codd Rules that state:

- For a database to be considered as a perfect relational database, it must follow the following rules:
- **Foundation Rule** - The database must be able to manage data in relational form.
- **Information Rule** - All data stored in the database must exist as a value of some table cell.
- **Guaranteed Access Rule** - Every unique data element should be accessible by only a combination of the table name, primary key value, and the column name.
- **Systematic Treatment of NULL values** - Database must support NULL values.
- **Active Online Catalog** - The organization of the database must exist in an online catalog that can be queried by authorized users.
- **Comprehensive Data Sub-Language Rule** - Database must support at least one language that supports: data definition, view definition, data manipulation, [integrity constraints](#), authorization, and transaction boundaries.
- **View Updating Rule** - All views should be theoretically and practically updatable by the system.
- **Relational Level Operation Rule** - The database must support high-level insertion, updation, and deletion operations.
- **Physical Data Independence Rule** - Data stored in the database must be independent of the applications that can access it i.e., the data stored in the database must not depend on any other data or an application.
- **Logical Data Independence Rule** - Any change in the logical representation of the data (structure of the tables) must not affect the user's view.
- **Integrity independence** - Changing the integrity constraints at the database level should not reflect any change at the application level.
- **Distribution independence** - The database must work properly even if the data is stored in multiple locations or is being used by multiple end-users.
- **Non-subversion Rule** - Accessing the data by low-level relational language should not be able to bypass the integrity rules and constraints expressed in the high-level relational language

Advantages of using the relational model

- The advantages and reasons due to which the relational model in DBMS is widely accepted as a standard are:
- **Simple and Easy To Use** - Storing data in tables is much easier to understand and implement as compared to other storage techniques.
- **Manageability** - Because of the independent nature of each relation in a relational database, it is easy to manipulate and manage. This improves the performance of the database.
- **Query capability** - With the introduction of [relational algebra](#), relational databases provide easy access to data via high-level query language like SQL.
- **Data integrity** - With the introduction and implementation of relational constraints, the relational model can maintain data integrity in the database.